VLSI · IECON 2026 CANDIDATE PAPER

An Activity-Rate-Aware Hysteresis Controller for Adaptive Power Gating

Dual-condition sleep-entry control for eliminating power-gating thrashing under bursty workloads

Joud Thaher · Layan Salem

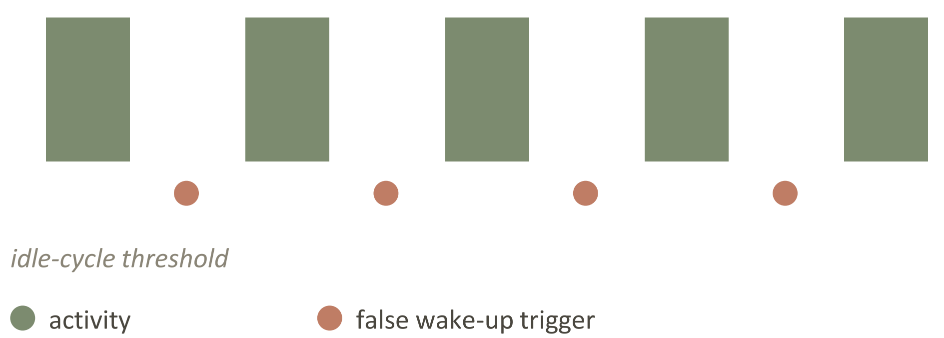
Supervised by Dr. Khader Mohammed

Department of Electrical and Computer Engineering — Birzeit University

Idle-cycle-count thresholds cannot distinguish genuine idleness from bursty activity

- Leakage power dominates total power dissipation as CMOS technology scales into the deep-submicron regime.
- Power gating is the standard leakage-reduction technique, but the sleep-entry policy is typically static: a fixed idle-cycle count triggers sleep, regardless of how the idle period arose.
- A workload of short bursts separated by gaps exceeding this fixed threshold causes repeated sleep/wake transitions — thrashing — even though the block is, in aggregate, far from idle.

A bursty workload as seen by a fixed threshold

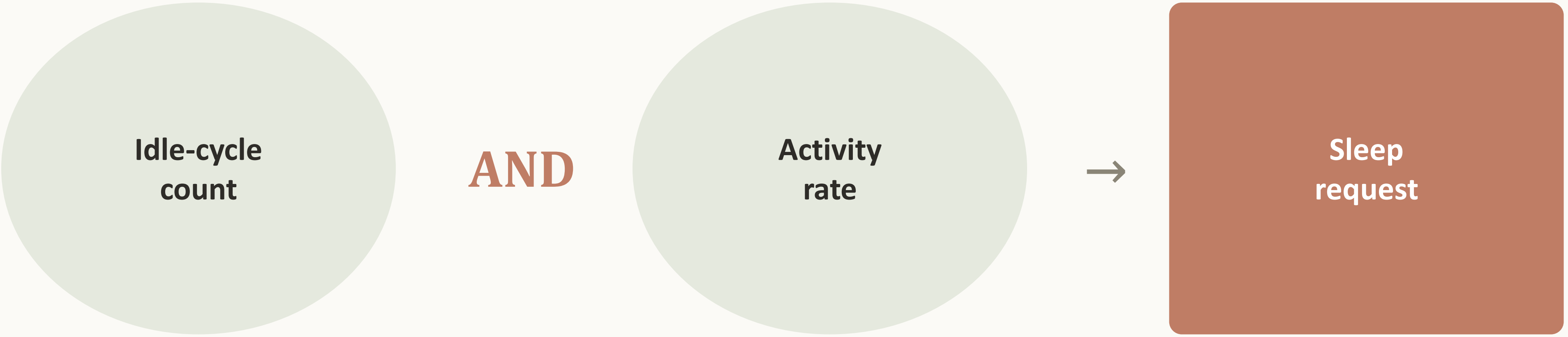


Each spurious wake-up costs latency and rush-current energy; the leakage saved by sleeping briefly can be outweighed by the cost of waking repeatedly.

Positioning relative to prior power-gating and adaptive power-management work

Reference	Approach	Key limitation	Gap addressed by this work
Agarwal et al. [3]	MTCMOS multi-level gating (fixed idle-cycle thresholds)	Static thresholds; no activity-rate check; thrashes under bursty gaps	Mode hierarchy retained; decision logic replaced with a dual-condition, rate-aware gate
Chang et al. [4]	Two-phase footer activation for rush-current control	Addresses only the wake-up mechanism, not entry decision	Wake-up mechanism adopted unchanged; contribution is orthogonal (entry, not transition)
Karthikeyan et al. [5]	ML-based monitoring, DVFS, core-level gating (multi-core)	Clustering/RNN hardware overhead hard to justify for one small block	Comparable adaptivity via fixed-function comparators — no learned model
Shiva & Patil [1]	Survey — DVS, power gating, clock gating	No new architecture proposed	Identifies lightweight, non-learned adaptive gating as an underexplored direction
Sahu et al. [2]	Design-optimization survey, low-power VLSI	No adaptive, rate-aware sleep-entry logic proposed	Supplies a concrete circuit-level instantiation of the identified gap
Tong et al. [6]	Lyapunov-based stability framework, IIoT task offloading	Different domain; not applicable at gate level	Motivates throughput preservation, not leakage alone, as an evaluation criterion

Sleep-entry gated on idle-cycle count AND activity rate



When recent activity is high, both entry thresholds widen — bursty traffic is no longer classified as idle.

- **Comparator-only logic** — 16-cycle shift-register popcount and simple comparators — no multipliers, no learned models.
- **Adaptive threshold widening** — Entry thresholds widen by a fixed offset when the activity rate exceeds a configurable ceiling.
- **Fair evaluation methodology** — Single-module substitution isolates the sleep-entry predictor as the sole experimental variable.

Two UPF power domains, six shared modules

PD_TOP — always-on control domain: activity monitor, hysteresis predictor, power-state FSM, sleep controller.
PD_ALU — power-gated domain: 32-bit ALU and MTCMOS footer switch.
The hysteresis predictor is the controller's only novel block.

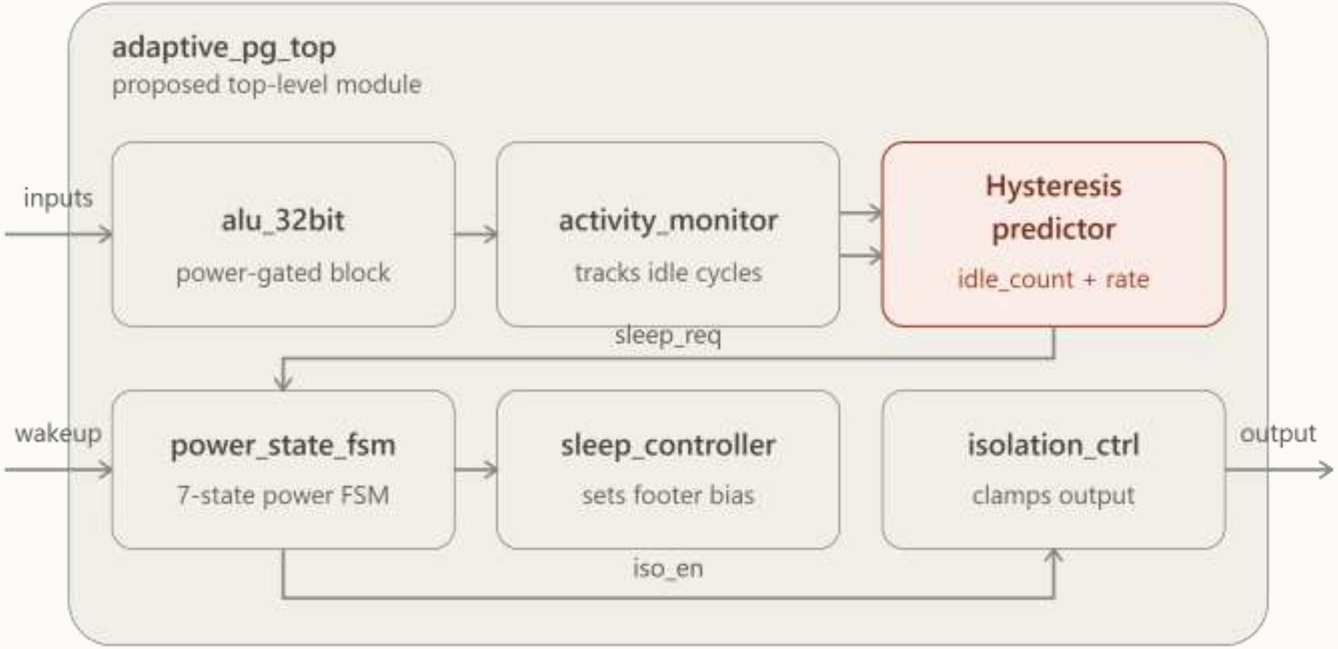


Fig. 1 — Architecture of the adaptive multi-level power-gating controller: two UPF power domains and inter-module signal flow.

Baseline vs. proposed: single-module substitution

Every block is bit-for-bit identical between the two designs except the sleep-entry predictor, isolating it as the sole experimental variable.

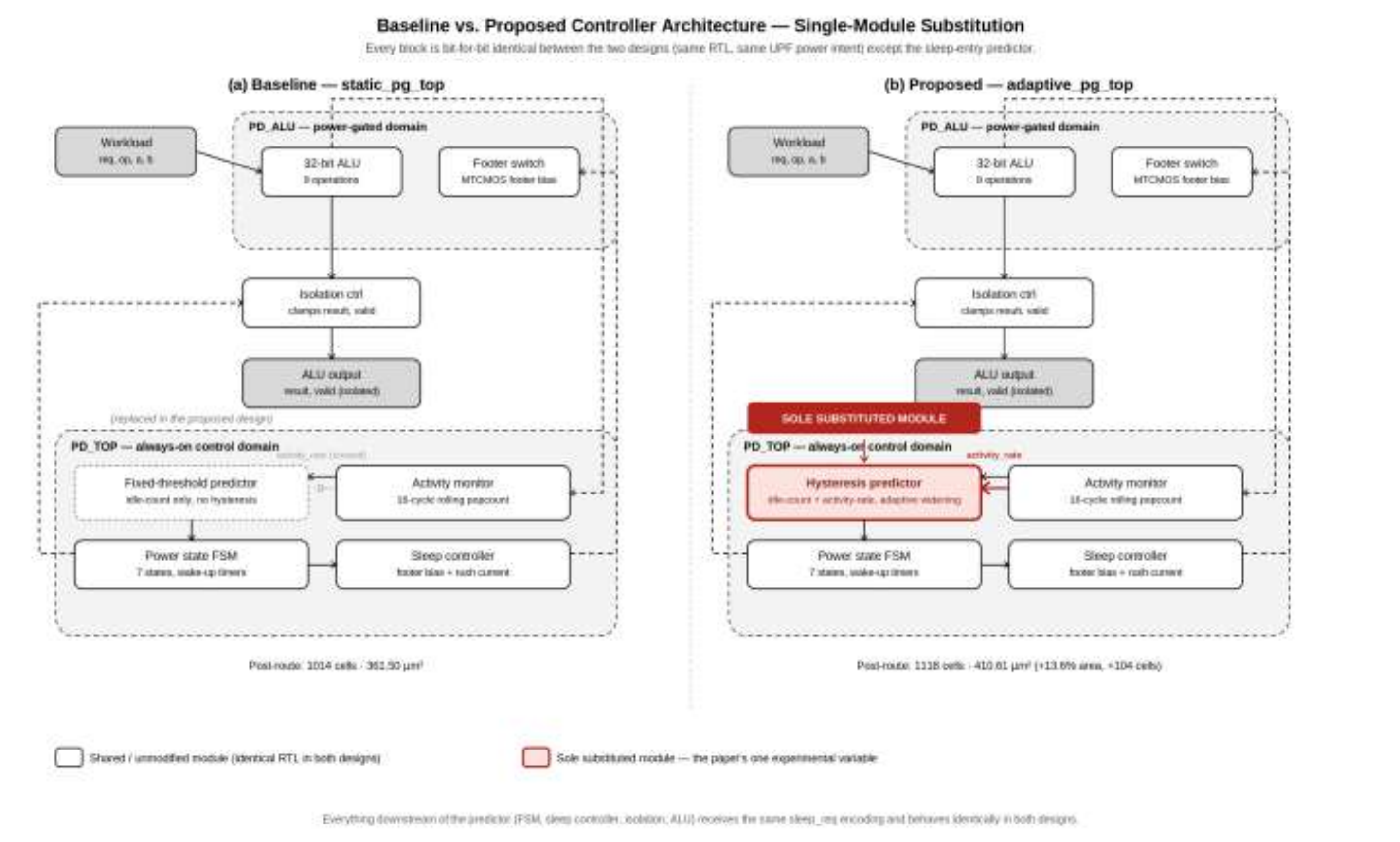


Fig. 2 — Baseline vs. proposed controller architecture under the single-module-substitution protocol (substituted block highlighted).

Hysteresis predictor: dual-condition sleep-entry gate

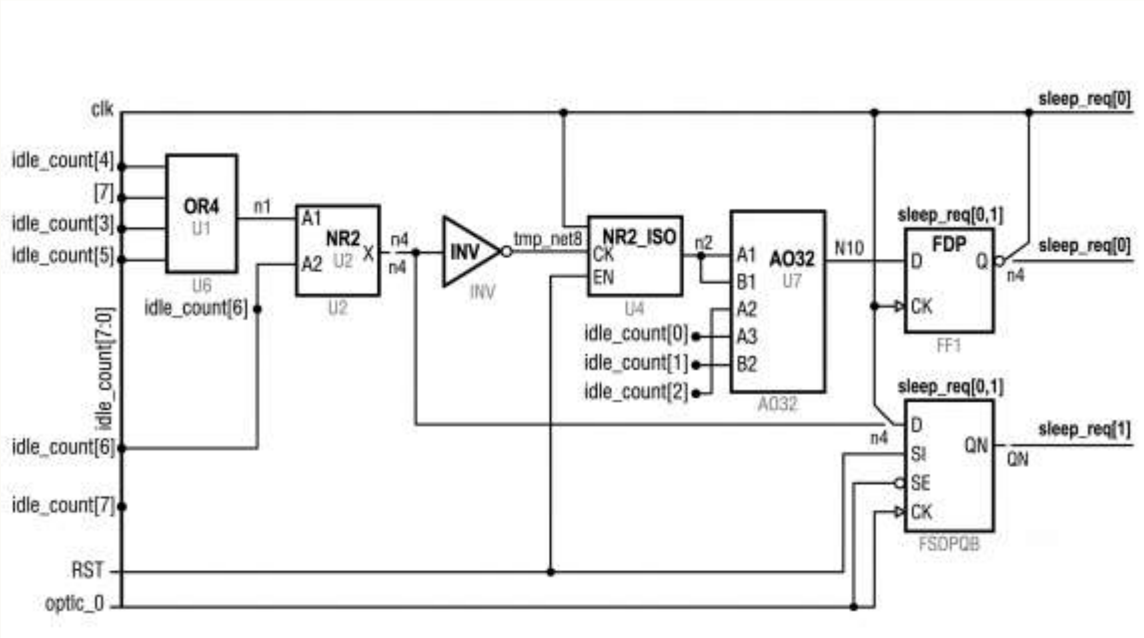
Activity monitor — 16-cycle rolling window implemented as a shift register; produces a consecutive idle-cycle count and an activity rate (population count of the window, scaled to 8 bits). No multipliers.

```
enter_LS = (idle_count > θ_LS) ∧ (rate < R_LS)      (1)

enter_DS = (idle_count > θ_DS) ∧ (rate < R_DS)      (2)

θ_i  =  θ_i,base + Δ        if  rate > T_high
θ_i  =  θ_i,base            otherwise      (3)
```

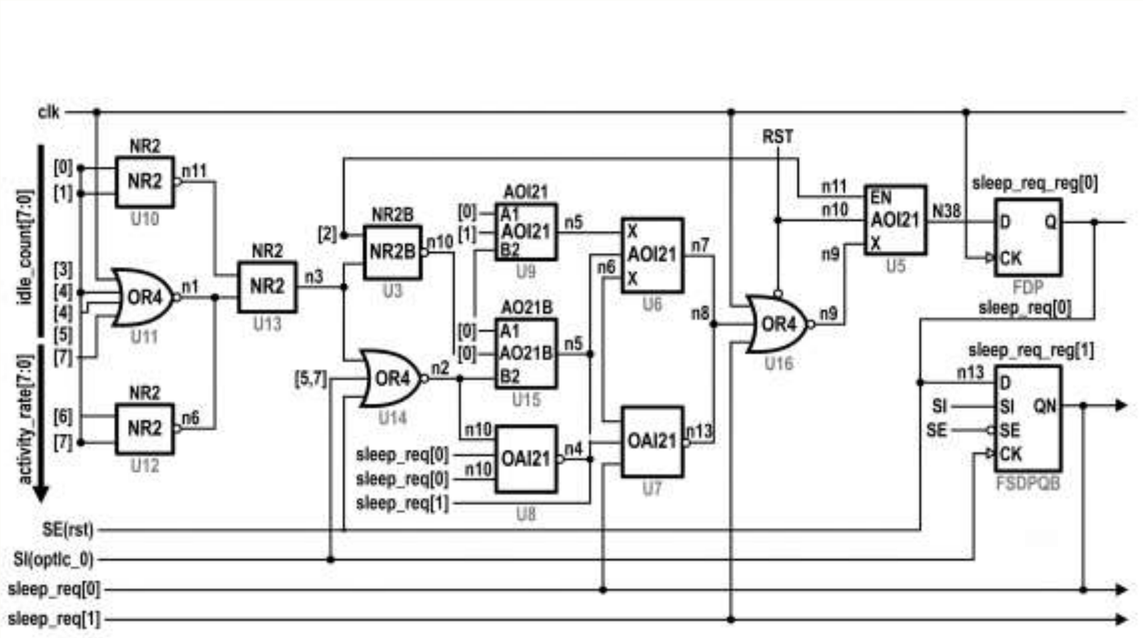
Base thresholds: $\theta_{LS,base} = 3$, $\theta_{DS,base} = 8$ cycles**Adaptive offset:** $\Delta = 6$ cycles**Exit thresholds:** $R_{LS} = 80$, $R_{DS} = 32$ (of 255)**High-activity trigger:** $T_{high} = 192$ (of 255)

Widening both thresholds when the activity rate exceeds T_high actively suppresses sleep entry during bursty-but-busy operation; narrower exit thresholds form the hysteresis band that prevents oscillation between states.

Synthesized gate-level schematics: baseline vs. proposed predictor



Baseline — idle_count[7:0] only (≈ 6 cells)



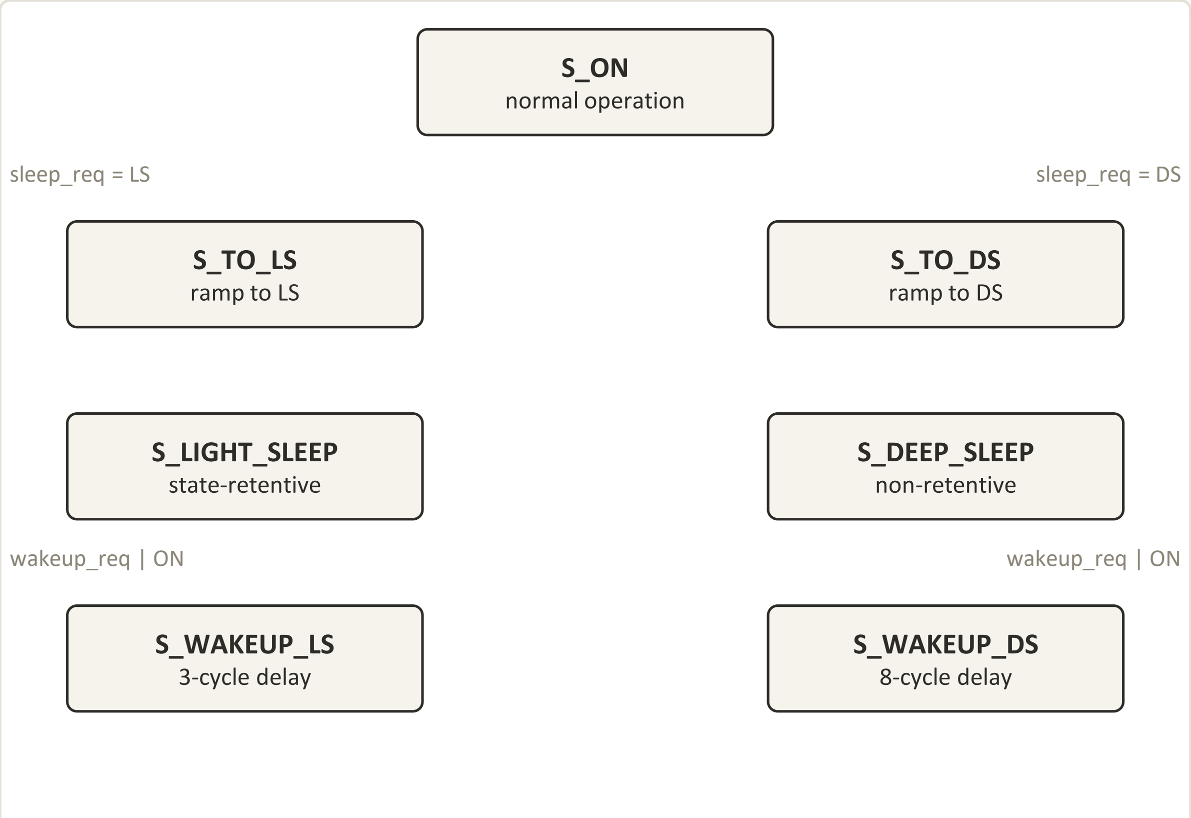
Proposed — idle_count[7:0] + activity_rate[7:0] (≈ 16 cells)

Figs. 3–4 (DC Ultra, SAED14LVT). Baseline: OR4NR2INVAO32 comparator chain. Proposed adds NR2, AOI21, OAI21, OR4 cells for the conjunction and hysteresis logic.

Power-state FSM: seven states govern every transition

3

cycles · Light Sleep wake-up delay



self-transitions omitted for clarity

8

cycles · Deep Sleep wake-up delay

Light Sleep retains state at ≈81% of ON-state leakage; Deep Sleep is non-retentive at ≈64% of ON-state leakage. Isolation asserts one cycle before sleep_n to prevent floating values from the gated domain.

One module swapped; everything else identical

Baseline

- 32-bit ALU
- Activity monitor
- Power-state FSM
- Sleep controller
- Isolation control
- UPF power intent

Fixed-threshold predictor

Proposed

- 32-bit ALU
- Activity monitor
- Power-state FSM
- Sleep controller
- Isolation control
- UPF power intent

Hysteresis predictor (novel)

only variable that differs

Complete RTL-to-GDSII flow

Four-stage Synopsys flow: RTL design → VCS simulation → DC synthesis → Fusion Compiler place-and-route

01 RTL Design

- 8 Verilog modules
- hysteresis_predictor ★ (novel)
- power_state_fsm — 7 states
- adaptive_pg_top / static_pg_top
- UPF: 2 power domains

02 VCS Simulation

- VCS T-2022.06-SP2
- 3 workloads: burst, sparse, mixed
- Proposed vs. baseline, identical stimulus
- 0 output mismatches
- 31.6% leakage saving (sparse)

03 DC Synthesis

- DC Ultra R-2020.09
- SAED14LVT, 14 nm
- LP: 5 ns (200 MHz) · HS: 2 ns (500 MHz)
- Both corners: timing MET
- Netlists + SDC written

04 Fusion Compiler P&R

- FC U-2022.12-SP5
- 30×30 µm² floorplan, 45.83% utilization
- 2,287 power/ground pins
- 0 placement violations, 0 open nets
- Post-route WNS +3.25 ns (MET)

Three synthetic workloads, identical stimulus, both designs

BURST

Frequent short bursts with idle gaps exceeding the baseline's fixed threshold, while duty cycle stays above the proposed design's activity-rate widening trigger.

SPARSE

Long, unambiguous idle periods — both controllers should converge to Deep Sleep with comparable leakage savings.

MIXED

Alternating dense and sparse phases, testing transition behavior under realistic variation.

Per-mode cycle counts, transition counts, wake-up events, and valid-result throughput were logged for each workload across both designs.

Synopsys VCS · Design Compiler · Fusion Compiler (IC Compiler II) · SAED14LVT, 14 nm

Per-mode cycle distribution and functional performance metrics

Table II — Per-mode cycle distribution across workloads (baseline vs. proposed)

Power state	Burst — Base.	Burst — Prop.	Sparse — Base.	Sparse — Prop.	Mixed — Base.	Mixed — Prop.
Cycles ON	78	135	8	16	92	104
Cycles Light Sleep	57	0	10	11	10	9
Cycles Deep Sleep	0	0	155	146	81	70
Total cycles	135	135	173	173	183	183

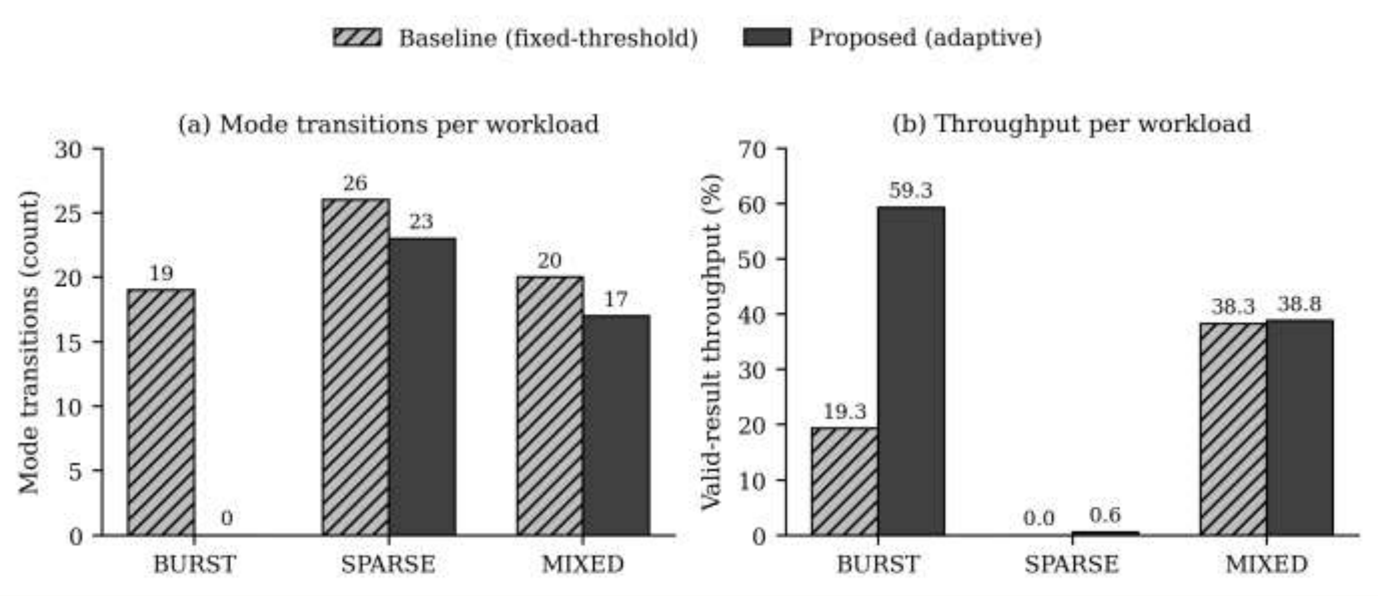
Table III — Functional performance metrics across workloads (baseline vs. proposed)

Metric	Burst — Base.	Burst — Prop.	Sparse — Base.	Sparse — Prop.	Mixed — Base.	Mixed — Prop.
Mode transitions	19	0	26	23	20	17
Wake-up events	9	0	8	7	7	6
Leakage saving (%)	8.8	0.7	33.4	31.6	17.0	14.7
Throughput (%)	19.3	59.3	0.0	0.6	38.3	38.8
Output mismatches	0	0	0	0	0	0

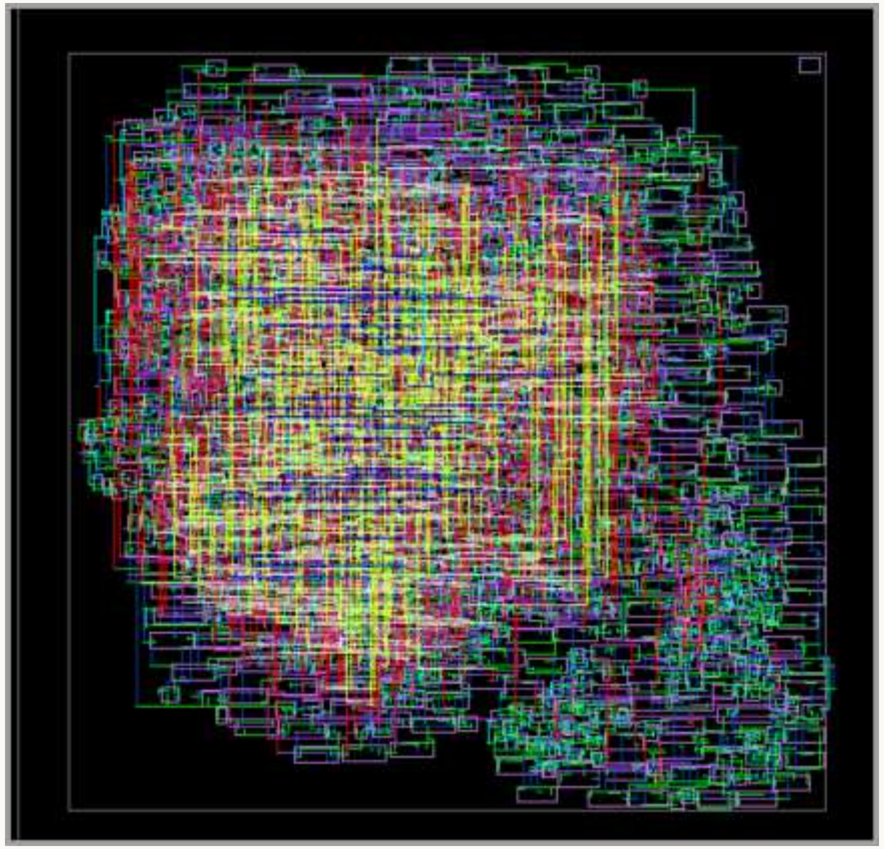
Zero output mismatches confirm the substitution introduces no functional regression in the shared datapath.

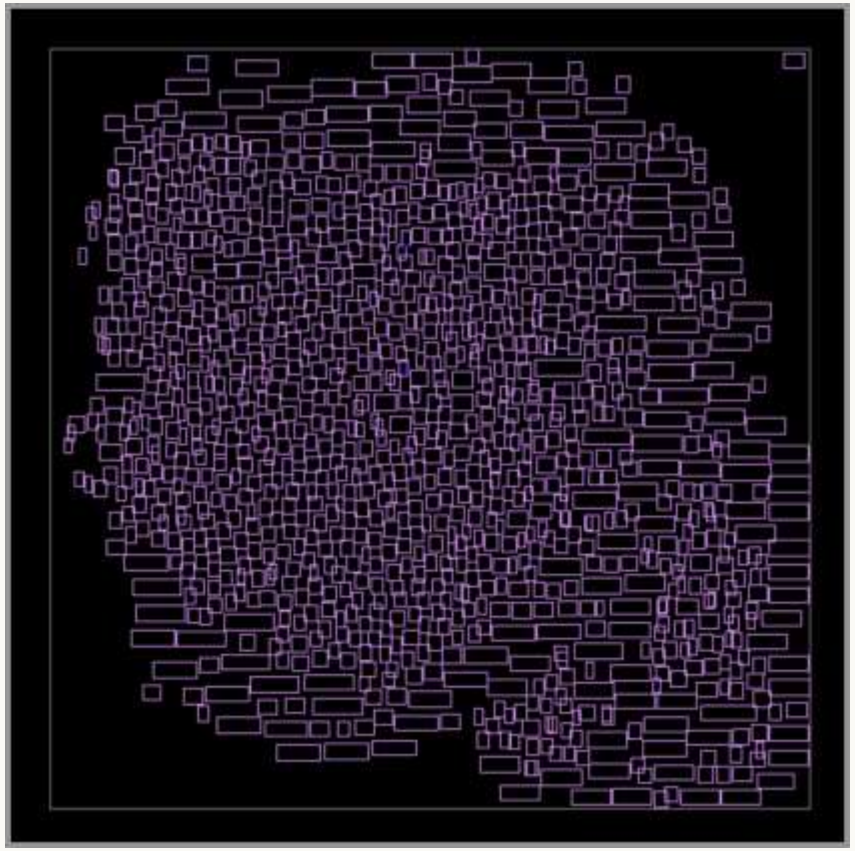
Zero mode transitions vs. nineteen, under burst traffic

VCS simulation waveform (SPARSE workload): power_mode and sleep_n, baseline vs. proposed.



Post-route layout: 30×30 μm² die, zero open nets





Die area: 30 × 30 μm²**Core utilization:** 45.83%**Standard cells:** 1,118 instances**Total wire length:** 9,873 μm**Placement violations:** 0**Open nets:** 0**Post-route WNS:** +3.25 ns (MET)

Fig. 7 — Fully routed layout (M1–M9 visible). *Fig. 8 — Cell placement view, 1,118 SAED14LVT instances at 45.83% utilization.*

Area and timing closure: both corners fully met

Table IV — Cell area summary (DC synthesis and post-route)

Design point	Baseline (µm²)	Proposed (µm²)	Overhead
DC synthesis, LP (5 ns)	350.98	422.51	+20.4%
DC synthesis, HS (2 ns)	351.25	442.76	+26.1%
Post-route (LP corner)	361.50	410.61	+13.6%
Post-route instances	1,014	1,118	+10.3%

Table V — Timing closure summary (DC synthesis and post-route)

Corner	Base. WNS	Base. TNS	Prop. WNS	Prop. TNS
DC, LP (5 ns)	+2.44 ns	0.00	+2.51 ns	0.00
DC, HS (2 ns)	+0.01 ns	0.00	+0.01 ns	0.00
Post-route, LP	+3.25 ns	0.00	+3.25 ns	0.00

Timing is fully closed — zero total negative slack (TNS) — at both the low-power (5 ns / 200 MHz) and high-speed (2 ns / 500 MHz) corners, for both designs, at synthesis and after place-and-route. **Post-route WNS improves to +3.25 ns** versus +2.51 ns at DC synthesis, reflecting shorter interconnect achieved by the Fusion Compiler router relative to the wireload estimate.

Real post-route area overhead (13.6%) is materially lower than the 20.4–26.1% suggested by pre-layout DC estimation.

A bounded cost for a real behavioral win

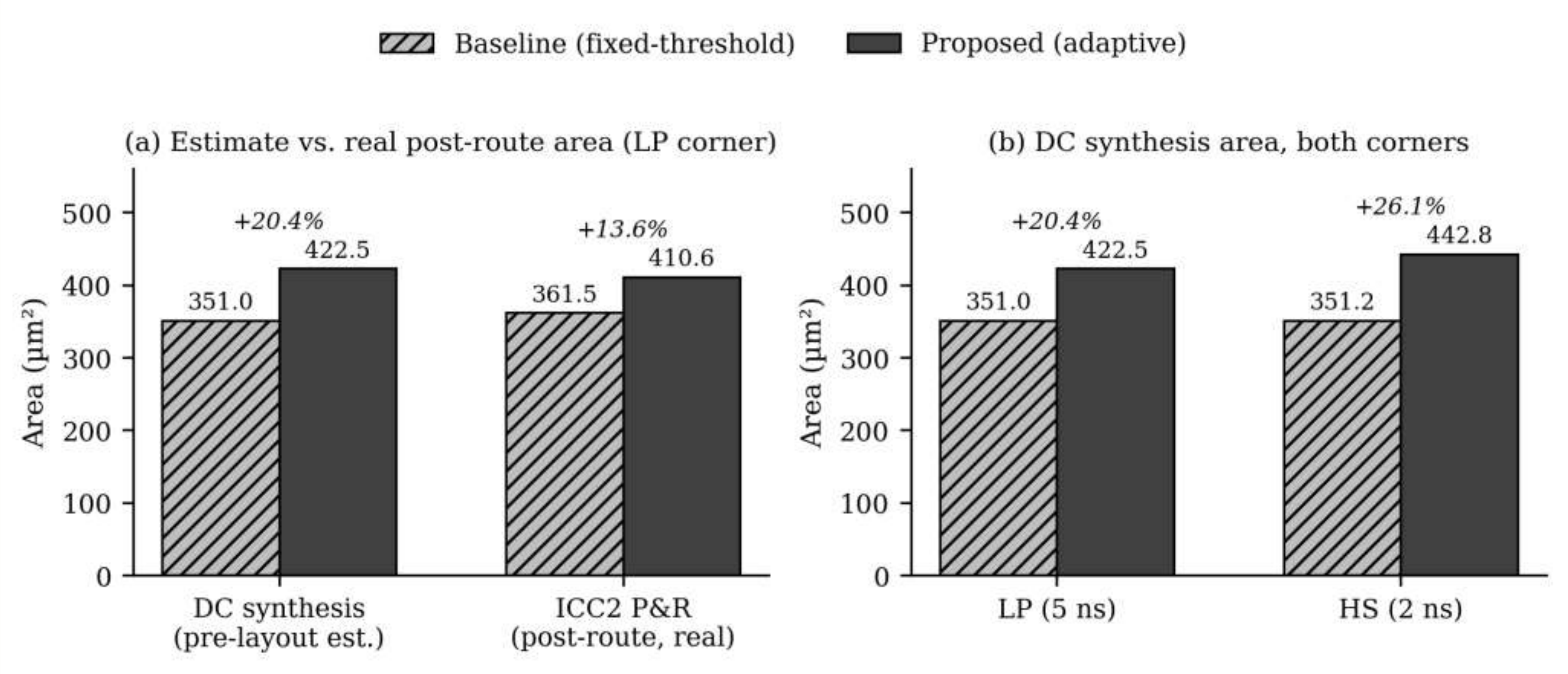


Fig. 10 — (a) Pre-layout estimate vs. real post-route area at the LP corner; (b) DC synthesis area at both corners, baseline vs. proposed.

Dynamic power and static leakage overhead

Table VI — Power summary, DC synthesis, LP corner (5 ns)

Power metric	Baseline	Proposed	Overhead
Cell internal power (µW)	7.09	13.54	+91.0%
Net switching power (µW)	4.71	5.64	+19.7%
Total dynamic power (µW)	11.80	19.18	+62.5%
Cell leakage power (nW)	627.2	991.3	+58.0%

Table VII — Post-route static leakage (LP corner)

Metric	Baseline	Proposed	Overhead
Static leakage (pW)	115,164	138,428	+20.2%
Runtime saving, SPARSE (%)	33.4	31.6	−1.8 pp
Runtime saving, MIXED (%)	17.0	14.7	−2.3 pp
Runtime saving, BURST (%)	8.8	0.7	−8.1 pp

Static leakage: full-power cost of the additional controller logic. **Runtime saving:** cycle-weighted leakage reduction during actual workload execution (simulation measurement) — the operational benefit.

The additional dynamic and leakage power reflect the always-on PD_TOP domain's extra combinational logic — the at-speed cost of the controller itself.

Where this sits against prior work

Approach	Decision input	ML hardware?	Outcome
Agarwal et al. [3] — multi-level gating	Idle-count only	No	Thrashes under bursts
Karthikeyan et al. [5] — multi-core ML	Learned classifier	Yes (ML)	Sized for multi-core, not one block
This work	**Idle-count + activity rate**	**No (comparators)**	**Bounded +13.6% area, zero thrashing**

vs. Agarwal et al. [3]: adds the activity-rate condition; eliminates thrashing under burst workloads (0 vs. 19 transitions).
vs. Karthikeyan et al. [5]: comparable qualitative adaptivity — suppresses sleep during bursts, sleeps during genuine idleness — with no inference hardware and no learned model.
vs. Chang et al. [4]: orthogonal contribution — the two-phase wake-up mechanism is adopted unchanged in both baseline and proposed designs.

What this work does not claim

- ! Energy comparisons use a derived metric (transitions × wake-up latency), not a first-principles joule measurement.
- ! Post-route results cover a single (LP) corner; multi-corner PVT analysis was outside this project's scope.
- ! Workloads are synthetic, designed to probe threshold-crossing behavior — not drawn from a real application trace.
- ! DRC verification was limited by the available PDK configuration; reported spacing violations reflect missing deck rules, not functional design errors.

Future work: extend the activity-rate gate to other block types, multi-corner verification, SPICE-level energy measurement, and validation against real application traces.

CONCLUSION

Comparator logic, real silicon, a bounded and quantified trade-off

3.1×	13.6%	0
throughput under burst traffic	real post-route area overhead	total negative slack — both corners, both designs

Results confirm the proposed design eliminates sleep-state thrashing while converging to baseline-level leakage savings under genuinely idle conditions, with timing fully closed at both the low-power and high-speed corners — at a real, bounded silicon cost.

Thank you.

Questions?

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Supervised by Dr. Khader Mohammed

Department of Electrical and Computer Engineering — Birzeit University